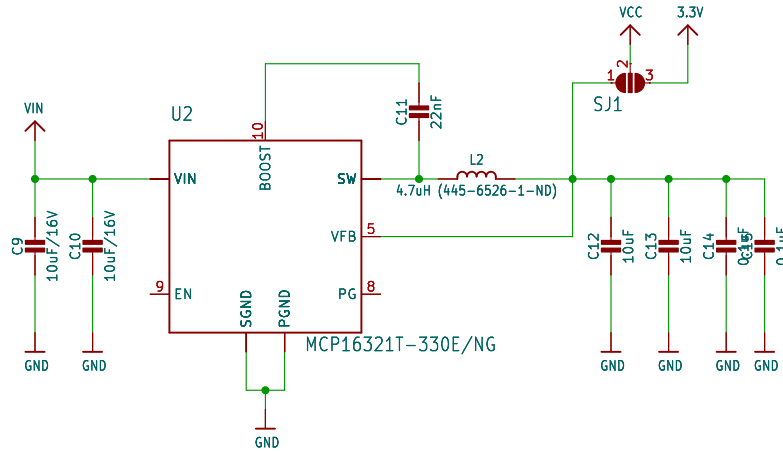


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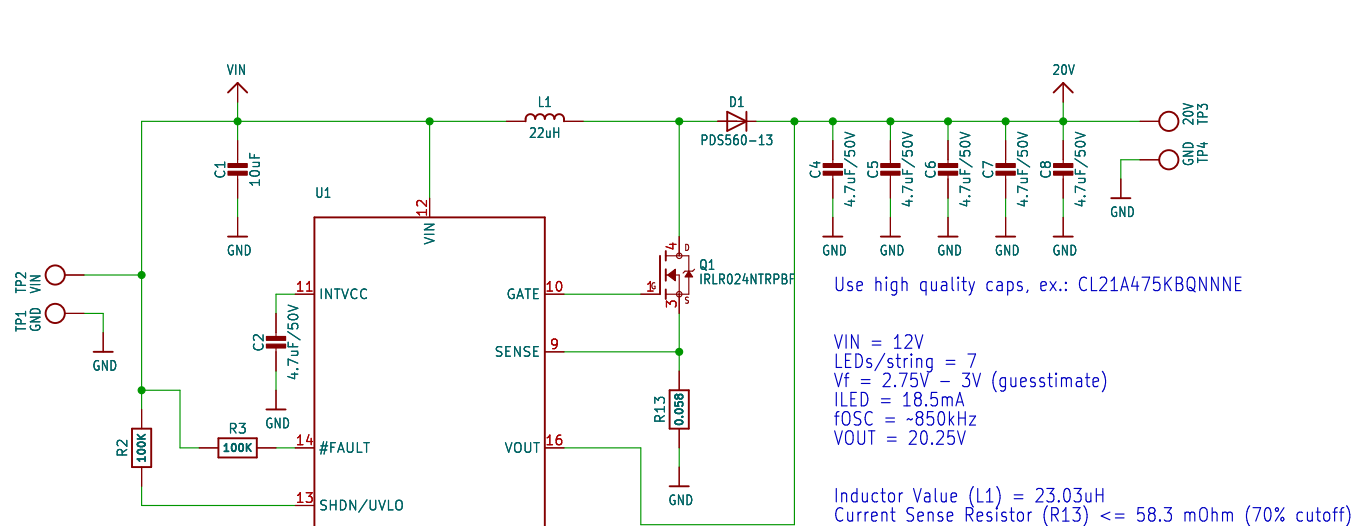
PROPRIETARY

### 3.3V Buck Converter

SJ1 Select buck or displayport for 3V3



### 20V Boost Converter



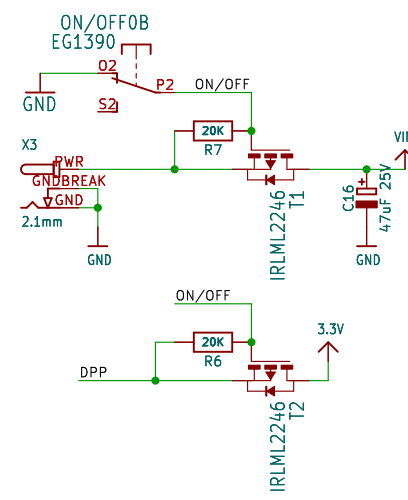
Use high quality caps, ex.: CL21A475KBQNNNE

VIN = 12V  
LEDs/string = 7  
Vf = 2.75V - 3V (guesstimate)  
ILED = 18.5mA  
fOSC = ~850kHz  
VOUT = 20.25V

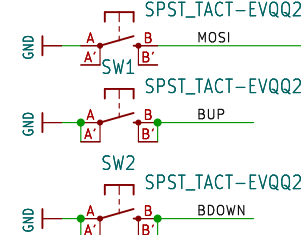
Inductor Value (L1) = 23.03uH  
Current Sense Resistor (R13) <= 58.3 mOhm (70% cutoff)

RT = 47K (850 KHz fOSC)  
RISET = 15K (~18.5mA LEDs)  
OVPSet = 47K + 20K (0.44V out)

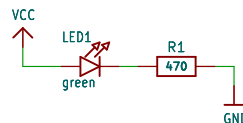
### On/Off Switch



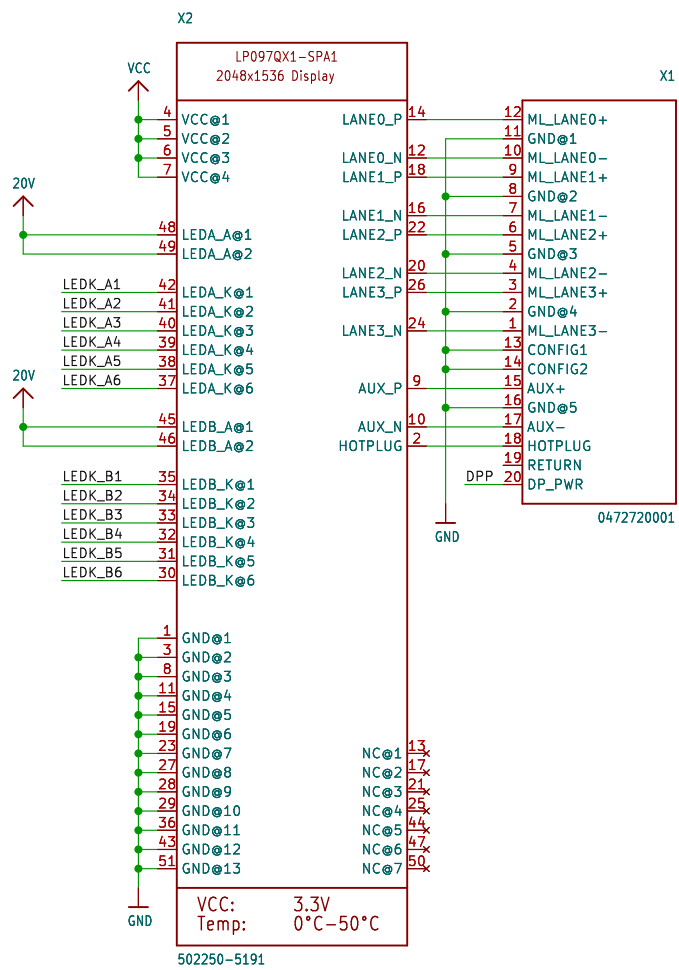
### Backlight Control



### Power Indicator



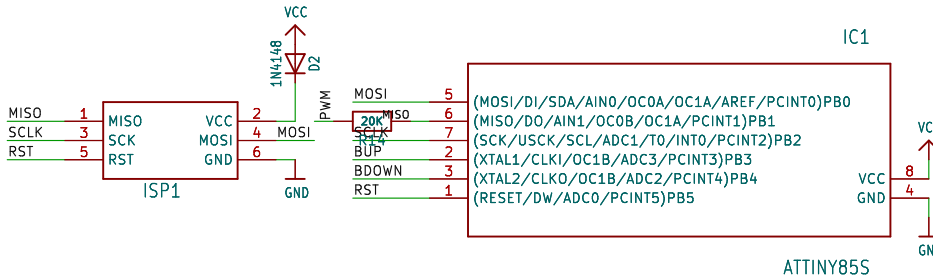
### Retina Display and DisplayPort Connector



Note: Sink side pinout used

Lanes are targetted at 85 Ohm differential (+/-15%)  
Avoid discontinuities in the reference plane  
DP\_PWR normally not connected

### MCU (for Backlight Control)



|              |                   |                  |                      |
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DisplayPort Layout Notes:  
[http://www.nxp.com/documents/application\\_note/AN10798.pdf](http://www.nxp.com/documents/application_note/AN10798.pdf)

|                                                  |       |         |   |
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